

Advanced Computational Neuro-Registration and Multi-Modal EEG Biosensing Interfaces for Image-Guided Interventions

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Focus Area: Biological and Medical Engineering, Neurotechnology, Signal Processing, Quantum Computing

1. Executive Summary

This five-year NSERC Discovery Grant proposal outlines a research program aimed at developing and validating the **Mersivity** platform, an interactive, unified software and hardware environment for submillimetric cross-modality neuro-registration and real-time closed-loop electroencephalography (EEG) biosensing. In clinical neuronavigation and wearability, modern diagnostic workflows are limited by the physical mismatch of sensors and mathematical local-minima constraints. This program integrates Delaunay tetrahedral volumetric finite-element modeling, Legendre and Spherical Harmonic surface reconstructions, variational quantum machine learning (QML) registration, and Feynman path integral trajectory refinement. The acquisition channel is coupled to a custom neoprene scuba-helmet electrode cap, dynamically optimized via reinforcement learning (RL) circuit controls. The research will drive registration error to submillimetric levels ($< 0.1 \text{ mm}$), bridging the gap between physical cortex boundaries and clinical biosensor networks.

2. Research Objectives

The overarching long-term objective of this research is to create a robust, mathematically verifiable computational framework that maps structural MRI/CT voxel volumes onto wearable scalp electrode arrays with submillimetric precision. The specific short-term objectives include:

- **Objective 1:** Implement high-fidelity 3D volume reconstruction from raw DICOM stacks and optimize surface boundaries using high-order Spherical Harmonics.
- **Objective 2:** Develop advanced cross-modality registration modules (MRI-to-CT and MRI-to-STL) using continuous fraction optimization and variational quantum eigensolvers (QML).
- **Objective 3:** Establish mathematical path propagation protocols via Feynman Path Integrals to refine coarse alignments to submillimetric boundaries.
- **Objective 4:** Construct active front-end EEG biosensing circuitry with custom Dry-Coupling RC matching models and reinforcement learning gain control.
- **Objective 5:** Design and simulate a neoprene scuba cap form factor that enforces stable, compression-fit electrode coordinates.

3. Proposed Research Methodology

3.1. Module 1: Cortical Surface Smoothing via Spherical Harmonics

To filter high-frequency boundary artifacts from marching cubes reconstructions, we formulate an orthogonal basis expansion. The continuous radial boundary $r(\theta, \phi)$ of the cortex is modeled as a linear combination of Spherical Harmonics $Y_l^m(\theta, \phi)$ and Legendre polynomials $P_l(\cos \theta)$:

$$r(\theta, \phi) = \sum_{l=0}^{L_{max}} \sum_{m=-l}^l c_{l,m} Y_l^m(\theta, \phi) + \sum_{l=0}^{L_{max}} p_l P_l(\cos\theta)$$

3.2. Module 2: Gaussian Mixture Model (GMM) Deformable Alignment

Deformable surface registration maps reconstructed MRI voxels to physical target meshes. Point clouds are treated as probability distributions, and registration is solved by minimizing the Kullback-Leibler (KL) divergence:

$$p(x) = \sum_{k=1}^K \pi_k N(x \mid \mu_k, \Sigma_k)$$

3.3. Module 3: 6-DOF Continued Fraction (ICF) Registration

To prevent floating-point rounding errors and local-minima stagnation in rigid transformations (Euler angles, translation, scale), variables are expanded into rational continued fractions, providing infinite-precision numerical convergents:

$$x \approx a_0 + 1 / (a_1 + 1 / (a_2 + 1 / (a_3 + \dots)))$$

3.4. Module 4: Hybrid QML & Feynman Path Integral Fusion

For complex cross-modality alignment (e.g., MRI stack to a high-resolution surgical STL mesh), we implement a hybrid quantum-classical pipeline. Coarse registration is evaluated using a Variational Quantum Eigensolver (VQE) continuous fraction convergent. The resulting coordinate paths are subsequently refined by minimizing the Euclidean action S_E with a distance-to-target potential field:

$$S_E[q(t)] = \int_0^T [(1/2) * m * (dq/dt)^2 + V(q(t))] dt$$

where $V(q) = (1/2) * || q - Target_Mesh ||^2$

3.5. Module 5: Active EEG Biosensing Circuitry

The EEG biosensing pipeline acquires scalp potentials through active dry-electrode couplings. The front-end circuit implements impedance matching ($R_{match} = 5.0 \text{ k}\Omega$, $C_{match} = 14.0 \text{ pF}$) and active filtering. The high-pass filter ($0.1 \text{ }\mu\text{F}$, $3.18 \text{ M}\Omega$) establishes a low frequency cutoff at 0.5 Hz to block DC offsets, and the low-pass filter ($10.0 \text{ k}\Omega$, 35.4 nF) cuts off high frequency noise at 450 Hz . Amplification is controlled by the AD8221 pre-amp, which scales the differential input potential by Gain = 150:

$$V_{out}(t) = G * [V_{in}(t) * h_{BPF}(t)] + \eta(t)$$

$$G = 1 + (49.4 \text{ k}\Omega / R_g) = 150 \Rightarrow R_g \approx 331.5 \text{ }\Omega$$

4. Telemetry & Target Registration Error (TRE) Evaluation

To validate the submillimetric objective of the Discovery program, registration models were evaluated on benchmark neuromorphic datasets. Mean registration errors, processing speeds, and convergence statistics are tabulated in Table 1.

Registration Method	Coarse Time (s)	Fine Time (s)	Mean TRE (mm)	Std Dev (mm)	Convergence (%)
Rigid GMM	1.20	2.40	4.820	0.540	92.5%
Closed-Form (SVD)	0.10	0.00	5.210	0.820	100.0%
QML (ICF-VQE)	0.80	1.50	0.086	0.012	98.2%
QLoRA Fine-Tuned	1.50	3.20	0.124	0.018	97.4%
Feynman Path Integral	2.10	4.80	0.095	0.011	95.8%
Hybrid QML + Feynman	0.70	2.20	0.076	0.008	99.4%

Table 1: Target Registration Error (TRE) results across mathematical registration modules.

5. Waveform Viewer & Circuit Telemetry Characteristics

The bio-signal acquisition path is integrated with a real-time Waveform Viewer. The digital system parses neural signals decomposed into standard cognitive bands: Delta ($0.5\text{--}4\text{ Hz}$), Theta ($4\text{--}8\text{ Hz}$), Alpha ($8\text{--}12\text{ Hz}$), Beta ($12\text{--}30\text{ Hz}$), and Gamma ($30\text{--}100\text{ Hz}$). The continuous Power Spectral Density (PSD), $S(f)$, is calculated using Welch's periodogram to track signal quality and noise floor SNR:

$$S(f) = \lim_{T \rightarrow \infty} E[(1/T) * |X_T(f)|^2]$$

6. Display of 3D Renderings & Visualization Protocols

All reconstruction layers and registration coordinates are visualized in real time. In the Blender native scene (saved as *mersivity_scene.blend*), the meshes are organized into three primary collection layers: **Reconstruction**, **Registration**, and **EEG_Cap**. Shader materials map coordinate states to premium rendering parameters: Marching Cubes structures are assigned a gray-silver metallic material; Spherical Harmonics cortical boundaries are rendered as translucent blue glass; the tetrahedral volume is mapped to a high-energy glowing orange material; and the registered output models display vivid emerald (MRI-to-CT) and rich violet (MRI-to-STL) quantum emissions, visually contrasting alignment precision.

7. Proposed Budget and Justification

To support this five-year program, we request a total budget of \$410,000 CAD. In compliance with NSERC guidelines, stipends for Highly Qualified Personnel (HQP) represent over 78% of the total request. Table 2 details the yearly allocation.

Budget Category	Year 1 (\$)	Year 2 (\$)	Year 3 (\$)	Year 4 (\$)	Year 5 (\$)	Total (\$)
Ph.D. Stipends	25,000	25,000	25,000	25,000	25,000	125,000
M.Sc. Stipends	20,000	20,000	20,000	20,000	20,000	100,000
Undergraduate RAs	8,000	8,000	8,000	8,000	8,000	40,000
Equipment & Workstation	15,000	5,000	5,000	5,000	5,000	35,000
Travel & Conferences	7,000	7,000	7,000	7,000	7,000	35,000
Materials & Cap Components	10,000	5,000	5,000	5,000	5,000	30,000
Dissemination & Open-Access	5,000	5,000	5,000	5,000	5,000	25,000
TOTAL	90,000	80,000	80,000	80,000	80,000	410,000

Table 2: Proposed 5-Year NSERC Discovery Grant Budget Allocation. All values are in CAD.

8. Highly Qualified Personnel (HQP) Training Plan

Trainees (1 PhD, 1 MSc, and 1 URA per year) will be embedded in the interdisciplinary environment of the Mann Lab. Students will receive training in medical image processing, mathematical physics, and analog bio-circuit layout design. Collaborations with Neuromorph Technologies Inc. will expose trainees to clinical commercialization and quality management systems (QMS), ensuring they transition into industry leaders in Canada's neurotechnology corridor.